

VEDANT PUNGLIYA

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EDUCATION

Iowa State University

Ames, IA

Bachelor of Science, Computer Science

Summa Cum Laude — GPA: 3.99/4.00 — Minors: Cyber Security Engineering, Data Science

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, JavaScript (Node.js)

AI/ML: PyTorch, TensorFlow, LLMs, RAG, LangChain, LangGraph, Vector Databases, Model Context Protocol

Cloud/DevOps: AWS (EC2, ECS, Fargate, S3, IAM, VPC), Docker, Jenkins, CI/CD, Terraform, Git

Backend/Web: Flask, FastAPI, Spring Boot, REST APIs, Microservices, Redis, RabbitMQ, MySQL

PROFESSIONAL EXPERIENCE

PrecisePK

San Diego, CA

Software & AI Engineer

May 2026 – Present

- Developing an antimicrobial stewardship platform to improve reliability and clinical decision support workflows.
- Building automated drug-disorder-pathogen validation checks for clinical data integrity.

Software & Algorithm Engineering Intern

Sep 2025 – Dec 2025

- **Boosted convergence stability 15%** via **FOCE-I algorithms** in Python/C++ for nonlinear PK/PD modeling.
- **Cut runtime 70%** via scalable **AWS pipeline** (ECS, CodeBuild, S3, DynamoDB) for parameter estimation.
- **Raised clinical accuracy** by optimizing 1/2/3 compartment model functions for complex pharma datasets.

2GoProducts

San Diego, CA

Generative AI & Software Engineering Intern

Jun 2025 – Sep 2025

- Built fault-tolerant multi-agent AI (1K+ daily reqs) via MCP-based AWS microservices with polyrepo CI/CD.
- **Hit 85% NL-to-SQL accuracy** via **RAG** pipeline with **FAISS** vector search and **SQLGlot** schema parsing.
- **Cut analyst workload by 60%** via multi-tenant GenAI slack-bot (SaaS).

Iowa State University

Ames, IA

Undergraduate AI/ML Research Scientist

Aug 2024 – Jun 2025

- **Investigated code reasoning** in LLMs through latent-space analysis and saliency-based interpretability studies.
- **Reduced insecure code generation** by developing **activation steering guardrails**.
- **Submitted** "Concept-Level Analysis of Internal Representations in Code Language Models" to Empirical Software Engineering (peer review, Apr 2026), studying latent-space interpretability in code LLMs.

Weekend AI

Pune, India

AI Engineering Intern

May 2024 – Aug 2024

- **Built 20+ AI proof-of-concept** systems using LangChain and CrewAI for workflow automation.

SELECTED PROJECTS

- **TEE-Crafter — Confidential Computing Orchestrator** — Tool that turns Docker applications into attested confidential services across AWS, Azure, and GCP, automating TEE deployment, provisioning, and verification.
- **BioSignify: ML E-Signature Platform** — Built a B2B platform using blockchain and vector databases; ready to process 100K+ authentications/day with 99.2% accuracy and sub-200ms latency.
- **Intelligent NLP-to-SQL Engine** — Developed a multi-tenant Transformer-based NL-to-SQL engine supporting 50K+ daily tenant queries through prompt-driven schema generation.

CERTIFICATIONS & HONORS

[AWS Certified Solutions Architect – Associate](#)

Nov 2025

Honors: Phi Beta Kappa, IEEE Eta Kappa Nu (HKN), Phi Kappa Phi, President's Excellence Scholarship

Leadership: CS Student Advisory Council; Workshop Lead – CS & Software Engineering Club